W-A-450/5
JANUARY 3, 1992

FEDERAL SPECIFICATION

DETECTOR, BEAMBREAK, INFRARED

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. The W-A-450C specification covers intrusion detection equipment which is deployed as a system to include, but is not to be limited to, detection of intrusions relating to protection of national security information, sensitive information and equipment, high dollar items, and weapons. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Unit, type and class. This specification covers Unit Number 1, Type V, Infrared Beambreak Detector. Detectors furnished under this specification shall fall into classes by having beam lengths as follows:

Class A - 10 m or less.
Class B - 25 m or less.
Class C - 100 m or less.

2. APPLICABLE DOCUMENTS.

The documents listed in W-A-450C/GEN, of the issues in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

3. REQUIREMENTS.

3.1 Qualification. The infrared beambreak detector furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list. It shall conform to the minimum specifications required herein and shall have been tested and shall have passed the qualification tests specified in paragraph 4.1.

3.2 Temperature and humidity ranges.

3.2.1 Operating range. All equipment shall be capable of

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operating within the temperature and humidity ranges indicated below:

0 degrees C to +49 degrees C and relative humidity of 0% to 85%, non-condensing.

All units shall withstand the tests in paragraphs 4.1.3 through 4.1.5. Additional requirements may be imposed by the procuring entity.

3.2.2 Nonoperating range. All equipment which is not powered up shall be capable of withstanding, without damage, the temperature range indicated below:

-46 degrees C to 71 degrees C.

3.3 Dependability. All equipment supplied under this specification shall have a minimum Mean Time Between Failure (MTBF) of 10 000 hours.

3.4 Operation. The infrared beambreak detector shall consist of a transmitter unit and a receiver unit with necessary control circuitry to create a line of intrusion detection with modulated infrared energy. An alarm shall not be initiated when the beam is attenuated 90% or less. Usable and published detection patterns shall be included in the manufacturer's literature. Failure to supply this information shall be cause for non-compliance of the sensor. The infrared beambreak detector shall, after initial setup, stabilize within five minutes after being powered on and after any alarm initiating stimulus is removed. An alarm shall be initiated if the beam is interrupted under the conditions stated in 4.1.1.2. The receiver shall be designed to resist defeat by the substitution of any other nonpulsed infrared light source. Input voltage shall be 8 to 18 volts DC, reverse polarity protected, or 10 to 28 volts AC, 60 cps. Single-pole, double-throw alarm switch or relay and separate single-pole, double-throw tamper switch or relay connections shall be provided at a terminal strip for connection to external alarm reporting equipment. Alarm and tamper switch impedance shall be as follows:

Open state:
- greater than 100 000 ohms.

Closed state:
- For solid state relays - less than 50 ohms.
- For mechanical relays - 0.1 ohms or less.

3.4.1 Additional capabilities. The manufacturer is encouraged to provide information in their technical literature

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concerning performance data that would exceed the minimum requirements of this specification.

3.5 Infrared transmitter and receiver assembly enclosures. The infrared transmitter and receiver assemblies shall be housed in enclosures having a removable cover arranged to provide ready access for mounting, wiring, alignment, and for sensitivity and other internal adjustments. Mounting holes shall be provided in the back of the enclosure in positions that are readily accessible (sufficiently displaced from all internal parts and the enclosure side) when all internal parts and wiring are in place. No external hardware or fasteners except the cover fasteners shall be used.

3.6 Beam alignment. The position of the transmitter and receiver modules inside the transmitter and receiver units shall be adjustable ±90 degrees horizontally and ±10 degrees vertically for surface mounted units and ±15 degrees horizontally and ±5 degrees vertically for flush mounted units. Module adjustment shall require removal of the enclosure cover.

3.7 Cover. The transmitter unit and receiver unit covers shall be designed such that any attempted access to the interior shall cause a tamper condition. The tamper circuits shall be in the secure state whenever the cover is in place and shall go to their alarm state when the cover is moved sufficiently to reveal the inside or to introduce a defeating device. The tamper switch shall be constructed of corrosion resistant materials and operate smoothly to prevent jamming.

3.8 Terminals. Screw terminals shall be provided for interconnecting wiring. Access to the terminals shall require removal of the enclosure cover.

3.9 Switch and/or relay assembly. Alarm and tamper switch and/or relay contacts shall be protected against environmental and excessive current conditions which may be damaging. Sustained currents up to 1.33 times the published current rating shall not cause damage to the switch or contacts.

3.10 Size, weight and power consumption. The size, weight and power consumption of each unit shall be the minimum practicable, consistent with meeting the requirements of this specification and shall be stated in the manufacturer's literature.

3.11 Maintainability. The infrared transmitter and receiver units shall be designed and constructed in such a manner as to minimize maintenance and repair time. To the maximum extent practicable, items shall be of a modular, unitized construction. Components shall be so arranged and assembled that

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they are readily visible and accessible to maintenance personnel. Specialized training is not required to maintain this equipment. Field maintenance shall consist of trouble shooting and isolation of failures to the device or replaceable sub-assembly and replacement of the defective device or sub-assembly. Preventive maintenance may be required. Adjustments shall require no disassembly of the detector beyond mere removal of the cover.

3.12 Life cycle. The infrared beambreak detector alarm and tamper switches and/or relays shall be rated at not fewer than 1 000 000 operations under a load of 250 milliampere driven from a 48 volt DC source. Contact resistance shall not change more than +20% from the initial reading.

3.13 EMI compatibility. Units shall operate normally when subjected to the testing of 4.1.6. The testing shall neither inhibit valid alarms nor generate invalid alarms.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Tests.

4.1.1 Performance test. The unit being tested shall be mounted, wired, and aligned in accordance with the manufacturer's recommendations in a room or area large enough to accommodate its published detection limit. The unit's alarm contacts shall be connected to a buzzer and its sensitivity shall be adjusted to the maximum limit for its class.

4.1.1.1 Alignment and detection limit test. If the manufacturer did not include a Beam Restricting Shield (BRS) made of opaque material with holes similar to the one in figure 1, a Neutral Density Filter (NDF), manufactured for use in the infrared wavelength range (0.75 to 300 microns) with a nominal density of 1.0, shall be used to check proper alignment.

Both the BRS and the NDF shall attenuate the infrared beam approximately 90%. To verify proper alignment the test engineer shall insert either the BRS or the NDF into the beam within 300 mm of the receiver and verify that an alarm is not initiated. If an alarm is initiated, realignment is mandatory.

If three additional attempts to align the detector are unsuccessful, the detector being tested shall be disqualified.

4.1.1.2 Beam interruption test. Once the unit is properly aligned, the test engineer shall test the proper operation of the detector by inserting a sheet of opaque material into the beam for about one second and verifying that an alarm was initiated.

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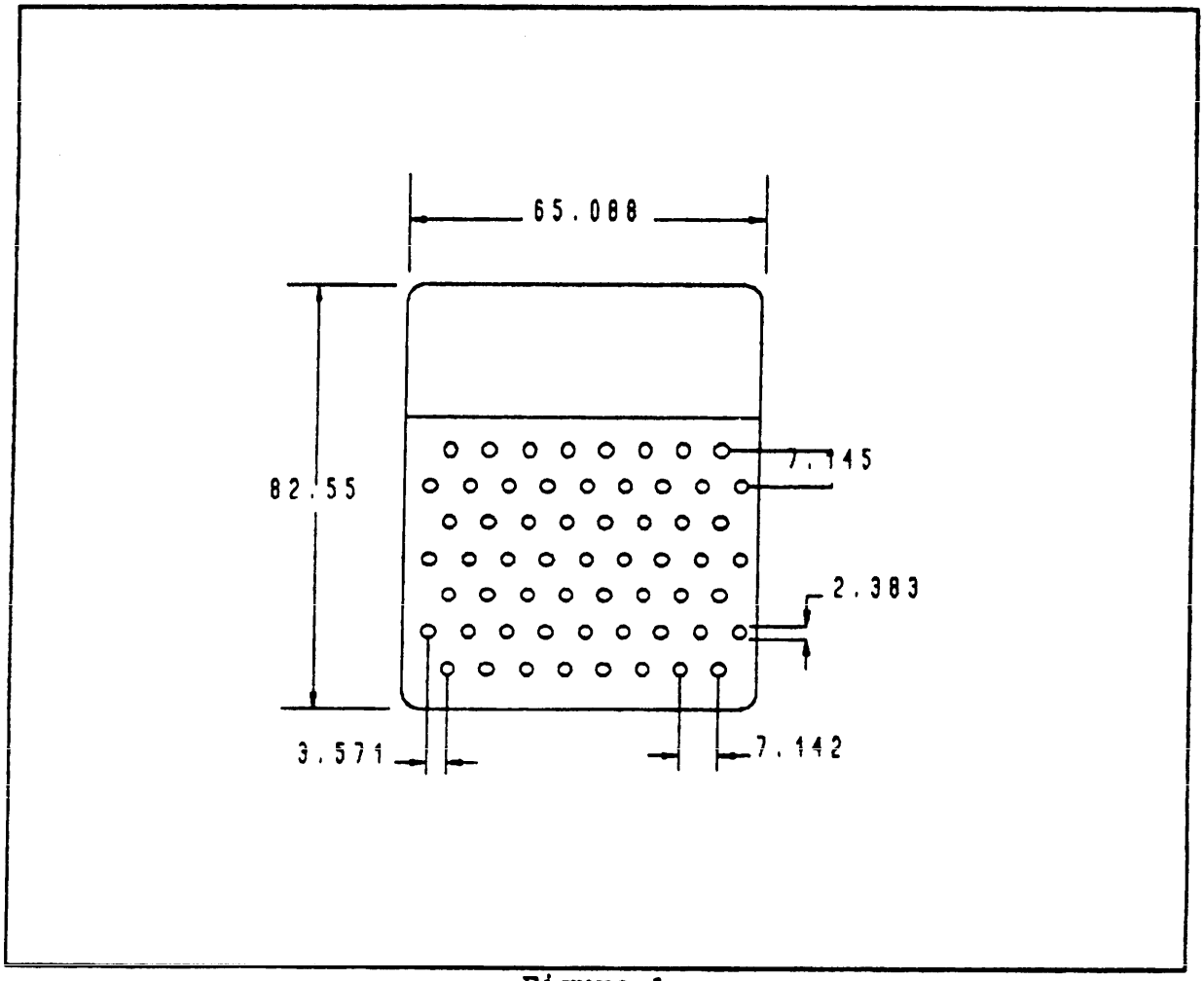


Figure 1
Beam Restricting Shield
(Dimensions in mm)

An alarm shall occur when the beam is interrupted by an object measuring 200 mm. in diameter and traveling at the speed of up to 0.3 m/s. The target shall be placed at the midpoint of the beam so that its position and direction of motion are perpendicular to the beam axis and the beam centerline.

4.1.1.3 Detection of attempt to capture receiver with foreign light source. The test engineer shall slowly direct an infrared light source into the receiver. An alarm shall be immediately initiated.

4.1.2 Stability test. The unit submitted for qualification testing shall perform its intended function in a stable

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environment free of stimuli that would initiate a nuisance alarm. The test period shall be 90 days of continuous operation with any adjusted controls set to provide the specified security. The test period shall commence after the unit has been installed in accordance with the manufacturer's instructions and after a "shakedown" operational period of not more than seven consecutive days.

4.1.3 High temperature test. Units submitted for qualification shall be energized for a period of four hours at 49 degrees C. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 49 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.4 Low temperature test. Units submitted for qualification shall be energized for a period of four hours at 0 degrees C. The unit shall then be tested as specified in paragraph 4.1.1 with the temperature maintained at 0 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.5 Humidity test. Units submitted for qualification shall be energized for a period of four hours with the humidity maintained at 85% at a maintained temperature of at least 26 degrees C. The test chamber shall be vented to the atmosphere to prevent the accumulation of moisture. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 26 degrees C and the humidity maintained at 85%. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature and humidity must be stabilized for 5 minutes before the test period commences.

4.1.6 EMI compatibility test. Units shall be tested for compliance with 3.13 in accordance with the applicable tests of MIL-STD-462 with consideration of the following paragraphs:

4.1.6.1 Test chamber. All testing shall be performed in an EMI test chamber.

4.1.6.2 Conducted electromagnetic susceptibility test. The units shall be subjected to power line conducted interference in accordance with test methods CS01, CS02, and CS06 of MIL-STD-462. Maintain a test level of 1 or 3 Volts RMS as required over the entire frequency range.

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4.1.6.3 Radiated electromagnetic susceptibility test. The units shall be subjected to radiation in accordance with test method RS03 of MIL-STD-462. Use an electric field of 1 Volt per meter for frequencies in the range of 14 Khz. to 2 Mhz. and 3 Volts per meter for frequencies in the range of 2 Mhz. to 12 Ghz.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking shall be in accordance with ASTM D3951. The level of packing shall be A, B, or C, as specified (see 6.2).

5.1.1 Civil agency marking. In addition to any special markings (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Sensor systems constructed with components qualified under this specification are intended to supplement or support protective personnel (e.g., guards) for areas requiring special security measures. These properly assembled systems are resistive to neutralization or to compromise by covert or surreptitious attack. Guidance in selecting a particular item for a particular security situation is not intended and is beyond the scope of this document. Qualification approval merely means that the item meets the specification requirements at the time it was tested for qualification.

6.2 Ordering data. The purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type of coverage (detection pattern) required (where applicable).
(c) Quantities required.
(d) Spare parts required.
(e) Destination inspection required.
(f) Levels of packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).
(h) Shipping schedule required.

6.3 Qualification. Regarding products which require qualification under this specification, awards will be made only for products which have been tested and approved for inclusion on the applicable Federal Qualified Products List prior to the time

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set for opening of bids whether or not such products have actually been listed. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is:

General Services Administration
Federal Supply Service, Furniture Commodity Center
Washington, DC. 20406

Information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

6.4.1 Alarms.

6.4.1.1 Valid alarm. A valid alarm is any alarm generated by stimuli which the detector is designed to recognize and which satisfies the detector's stated alarm criteria.

6.4.1.2 Invalid alarm. Any other alarm.

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